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Name: _____

May 11, 2013 Saturday

Department/School: _____

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University Physics
Midterm Examination

Kuang Yaming Honors School, Nanjing University

Select five out of following six problems.

1. (20pts) A particle moves in a one-dimensional space. The potential energy of the particle is given by

$$U(x) = 6x^2 - 2x^3.$$

- Determine the force acting on the particle.
- At what positions is the particle in equilibrium?
- Which of these equilibrium positions are stable and which are unstable?

Solution: (a) $F(x) = -\frac{dU}{dx} = -12x + 6x^2$

(b) From $F(x) = -\frac{dU}{dx} = -12x + 6x^2 = 0$, we have two equilibrium position $x = 0$ and $x = 2$.

(c) $U''(x) = 12 - 12x$. Since $U''(0) > 0$, then $x = 0$ is a stable equilibrium; since $U''(2) < 0$, then $x = 2$ is an unstable equilibrium.

2. (20pts) A ball of mass m moves under the uniform gravity and in a medium that exerts a resistance force $f_r = Cv^4$ on it. When the ball falls vertically, its terminal speed is v_0 .

- Express the coefficient C in terms of gravity g , mass m and terminal speed v_0 .
- Write the equation of motion for the upward motion of the ball.
- If the ball is projected vertically upward with an initial speed u , find the maximum height the ball can reach in terms of g , v_0 and u .
- Show that, however large the initial speed u may be, the maximum height the ball can reach is always less than $\frac{\pi v_0^2}{4g}$.

(Hint: $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$)

Solution: (a) $mg - Cv_0^4 = 0$. $C = \frac{mg}{v_0^4}$.

(b) $m \frac{dv}{dt} = -mg - Cv^4$ or $\frac{dv}{dt} = -g \left(1 + \frac{v^4}{v_0^4}\right)$

(c) Noting that $\frac{dv}{dt} = \frac{dv}{dz} \frac{dz}{dt} = v \frac{dv}{dz}$, the equation of motion can be rewritten as

$$v \frac{dv}{dz} = -g \left(1 + \frac{v^4}{v_0^4}\right) \quad \text{or} \quad \frac{v dv}{v^4 + v_0^4} = -\frac{g}{v_0^4} dz \quad \text{or} \quad \frac{dv^2}{v^4 + v_0^4} = -\frac{2g}{v_0^4} dz.$$

Integrating both side of the equation, we have

$$\int_u^0 \frac{dv^2}{v^4 + v_0^4} = -\frac{2g}{v_0^2} \int_0^h dz \quad \text{or} \quad -\frac{1}{v_0^2} \tan^{-1}\left(\frac{u^2}{v_0^2}\right) = -\frac{2g}{v_0^4} h \quad \text{or} \quad h = \frac{v_0^2}{2g} \tan^{-1}\left(\frac{u^2}{v_0^2}\right).$$

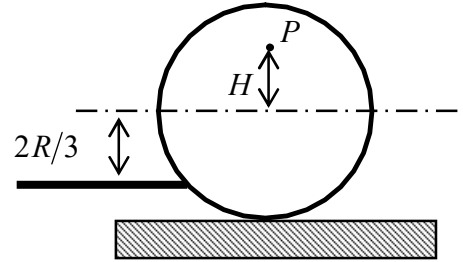
Because $\tan^{-1}\left(\frac{u^2}{v_0^2}\right) < \frac{\pi}{2}$, therefore $h < \frac{\pi v_0^2}{4g}$.

3. (20pts) A billiard ball of radius R is given a sharp horizontal blow by a cue stick. The hitting point is $2R/3$ below the centerline of the ball. The ball is at rest on the table initially and it obtains a speed v_0 just after the hitting. The coefficient of kinetic friction between the ball and the table is μ_k .

(a) What is the angular speed ω_0 of the ball just after the hitting?

(b) A point P that is at height H above the center of the ball is at rest relative to the table just after the hitting. Find H .

(c) What is the speed of the ball once it begins to roll without slipping?



Solution: (a) The impulse is mv_0 and the rotational impulse is

$$I_\tau = mv_0 \times \frac{2}{3}R = \frac{2}{3}mv_0R. \quad \text{From } I_\tau = I\omega_0 = \frac{2}{5}mR^2\omega_0, \text{ we have } \omega_0 = \frac{5}{3}\frac{v_0}{R} \text{ (The spin is backward).}$$

(b) From $v_P = v_0 - \omega_0 H = v_0 - \frac{5H}{3R}v_0 = 0$, we have $H = \frac{3}{5}R$.

(c) The equation of translational motion before pure rolling is

$$m \frac{dv}{dt} = -mg\mu_k \quad \text{or} \quad dv = -\mu_k g dt.$$

Integrating both sides of the equation, we have

$$\int_{v_0}^v dv = -\int_0^t \mu_k g dt \quad \text{or} \quad v = v_0 - \mu_k g t.$$

And the equation of rotational motion is

$$I \frac{d\omega}{dt} = mg\mu_k R \quad \text{or} \quad d\omega = \frac{5}{2R} \mu_k g dt.$$

Integrating both sides of the equation, we have

$$\int_{-\omega_0}^{\omega} d\omega = -\frac{5}{2R} \mu_k g \int_0^t dt \quad \text{or} \quad \omega = -\omega_0 + \frac{5}{2R} \mu_k g t.$$

When $v = \omega R$, it begins to roll without slipping.

$$v_0 - \mu_k g t = \left(-\omega_0 + \frac{5}{2R} \mu_k g t\right)R = -\frac{5}{3}v_0 + \frac{5}{2} \mu_k g t.$$

Solving the equation above, we have $t = \frac{16v_0}{21\mu_k g}$.

Therefore, $v = v_0 - \mu_k g t = \frac{5}{21}v_0$.

4. (20pts) It is observed that the minimum distance of a comet of mass m_c from the sun is exactly the half of the radius of the earth's orbit around the sun, and its speed at that point is twice the orbital speed of the earth. Ignore the effects of the earth and other planets on the comet, assume the mass of the comet and the mass of the earth is negligible compared to the mass of the sun M_s , and assume the earth's orbit is circular with radius R and in the same plane of the comet's orbit.

(a) Find the orbital speed of the earth v_E in terms of the mass of the sun M_s , the radius of its orbit R and the gravitational constant G .

(b) What is the total energy of the comet? What type of orbit the comet travels in around the sun?

(c) Find the speed of the comet when it crosses the earth's orbit.

(d) Find the angle of the angle between the comet's orbit and the earth's orbit at which they cross.

Solution: (a) $m_e \frac{v_E^2}{R} = G \frac{M_s m_e}{R^2}, v_E = \sqrt{\frac{GM_s}{R}}.$

(b) The potential energy is $U = -G \frac{M_s m_c}{\frac{R}{2}} = -2 \frac{GM_s m_c}{R}.$

The kinetic energy is $T = \frac{1}{2} m_c (2v_E)^2 = 2m_c v_E^2 = 2 \frac{GM_s m_c}{R}.$

The total energy is $E = T + U = 0.$

The orbit of the comet is a parabola.

(c) The conservation of mechanical energy $\frac{1}{2} m_c v^2 - \frac{GM_s m_c}{R} = 0.$

$$v = \sqrt{\frac{2GM_s}{R}} = \sqrt{2}v_E.$$

(d) The conservation of angular momentum: $m_c (2v_E) \frac{R}{2} = m (\sqrt{2}v_E) R \sin \theta$

where θ is the angle between the velocity of the comet and the position vector starting from the sun.

$$\sin \theta = \frac{\sqrt{2}}{2}, \theta = \frac{\pi}{4}.$$

The angle between the comet's orbit and the earth's orbit $\alpha = \frac{\pi}{2} - \theta = \frac{\pi}{4}.$

5. (20 pts.) A stick of length L is initially standing vertically on the floor. If the stick falls over, with what angular velocity will it hit the floor? Assume that the end in contact with the floor does not slip.

Solution: Conservation of mechanic energy

$$\frac{1}{2} mgL = \frac{1}{2} I \omega^2, I = \frac{1}{3} mL^2.$$

$$\omega^2 = \frac{3g}{L}$$

$$\omega = \sqrt{\frac{3g}{L}}.$$

6. (20pts) As shown in the figure, a thin square plate of mass m is suspended vertically at one of its corners in a way so that it can oscillate freely (left and right) in the plane of the square. The side length of the square is a .

(a) Write down the equation of rotational motion of the square plate.

(b) Find the period of small oscillations of the plate.

(c) If the square plate oscillates (back and forth) in the direction perpendicular to its plane, what is the period of small oscillations?

Solution: (a) The rotational inertia about the suspension point is

$$I = I_c + md^2 = \frac{1}{6}ma^2 + m\left(\frac{\sqrt{2}}{2}a\right)^2 = \frac{2}{3}ma^2.$$

Thus, the equation rotational motion is

$$I\ddot{\theta} = -mg \frac{\sqrt{2}}{2} a \sin \theta \quad \text{or} \quad \ddot{\theta} + \frac{3\sqrt{2}}{4} \frac{g}{a} \sin \theta = 0.$$

(b) For small oscillations, we have $\sin \theta \approx \theta$. The equation of motion can be rewritten as

$$\ddot{\theta} + \frac{3\sqrt{2}}{4} \frac{g}{a} \theta = 0.$$

Therefore, $\omega^2 = \frac{3\sqrt{2}}{4} \frac{g}{a}$, $T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{4}{3\sqrt{2}} \frac{a}{g}}.$

(c) The rotational inertia about the suspension point is

$$I' = I'_c + md^2 = \frac{1}{12}ma^2 + m\left(\frac{\sqrt{2}}{2}a\right)^2 = \frac{7}{12}ma^2.$$

Thus, the equation rotational motion is

$$I'\ddot{\theta} = -mg \frac{\sqrt{2}}{2} a \sin \theta \quad \text{or} \quad \ddot{\theta} + \frac{6\sqrt{2}}{7} \frac{g}{a} \sin \theta = 0..$$

For small oscillations, we have $\sin \theta \approx \theta$. The equation of motion can be rewritten as

$$\ddot{\theta} + \frac{6\sqrt{2}}{7} \frac{g}{a} \theta = 0.$$

Therefore, $\omega^2 = \frac{6\sqrt{2}}{7} \frac{g}{a}$, $T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{7}{6\sqrt{2}} \frac{a}{g}}.$

